Model Card: Closed-Domain Retrieval-Augmented Generation (RAG) System

1. Performance Metrics

Variant B demonstrated superior performance across key evaluation metrics. Correctness improved from 79.2% to 88.0%, groundedness increased from 83.2% to 92.0%, and hallucination rate decreased from 8.4% to 3.6%. Latency increased moderately from 2.10s to 2.58s, remaining within acceptable operational thresholds.

2. Training Data Description

The system operates on a closed-domain document corpus indexed using FAISS. Documents are embedded using a pre-trained embedding model and retrieved at query time. No additional fine-tuning of the base language model was performed.

3. Limitations and Failure Modes

The system is constrained to its document corpus and cannot generalize to unseen domains. Failure modes include incorrect retrieval leading to inaccurate answers, incomplete context retrieval, and increased latency under heavier retrieval configurations.

4. Ethical Risks and Considerations

Risks include potential misinformation if retrieval fails, lack of transparency in retrieval ranking, and outdated or incomplete knowledge in the document corpus. The system avoids personal data usage, reducing privacy concerns.

5. Intended Use and Out-of-Scope Applications

Intended use includes domain-specific question answering where reliability is critical. Out-of-scope applications include open-domain general knowledge queries, real-time decision-making, and safety-critical systems.

Lineage Diagram

Knowledge Base → Ingestion Pipeline → Chunking & Metadata → Embeddings → Vector Index (FAISS) User Query → Preprocessing → Retrieval → Context + Query → Prompt → LLM Inference → Response + Attribution

Audit Trail

Timestamp	Event	Description	Action
T1	Model Update	Transition from Variant A to Variant B	Approved deployment

T2	Evaluation	A/B test statistical validation completed	Decision logged
T3	Monitoring	Latency and request metrics tracked	Continuous monitoring enabled

Risk register

Risk ID	Category	Risk Description	Trigger Condition	Mitigation Strategy
R1	Bias	Uneven or incomplete knowledge base may lead to biased or skewed responses	Repeated incorrect or one-sided answers detected	Expand and diversify knowledge base; periodic audits
R2	Robustness	System fails or produces incorrect answers for ambiguous or out-of-scope queries	Increase in errors or low-confidence responses	Implement fallback responses and refusal logic
R3	Privacy	Sensitive or confidential data may be included in the knowledge base	Detection of PII or restricted content in corpus	Data sanitization, access control, and document filtering
R4	Compliance	System may violate internal or regulatory policies (e.g., data governance)	Audit failure or policy breach detected	Enforce governance policies and compliance checks